Day 7 Task: Linux Terminal Lab

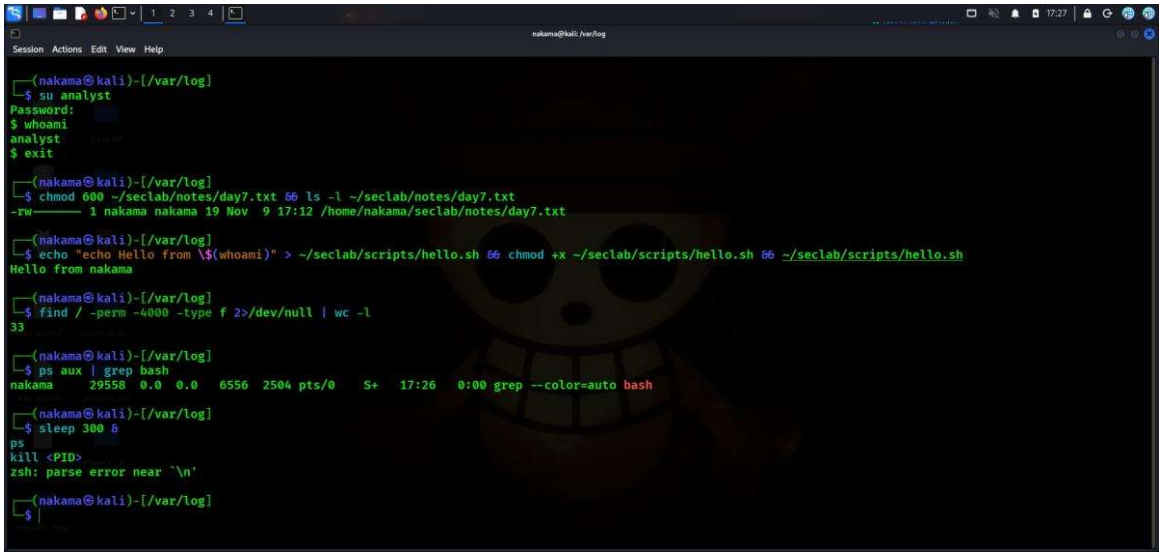
Step 1: The 20 tasks

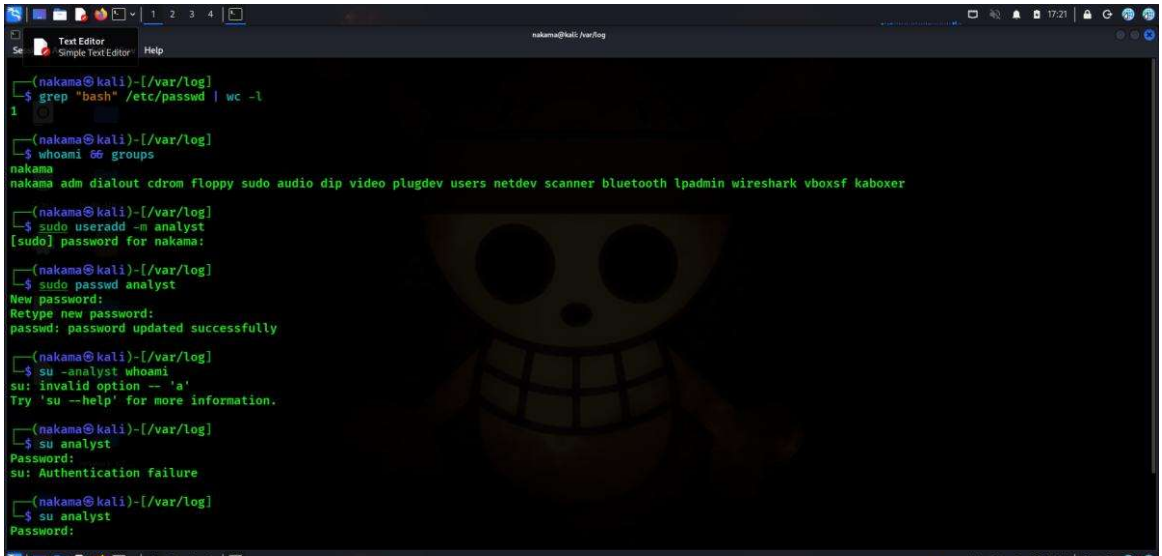
#	Area	Command(s) you used	Result/note
1	Navigation	pwd , ls -al	Outputs current directory (/home/nakama) and a detailed list including hidden files.
2	Navigation	cd /var/log , ls -lt | head -n 6	Moved to /var/log and showed the 5 most recently modified files.
3	Files	mkdir -p ~/seclab/notes ~/seclab/scripts	Created both folders simultaneously without error.
4	Files	echo "Linux lab started." > ~/seclab/notes/day7.txt && cat ~/seclab/notes/day7.txt	Created file and printed "Linux lab started." to terminal.
5	Files	cp ~/seclab/notes/day7.txt ~/seclab/notes/backup.txt && mv ~/seclab/notes/backup.txt ~/seclab/notes/day7-backup.txt	Successfully copied and then renamed the file.
6	Search	find /etc -type f -name "*.conf" | wc -l	Counted the .conf files and outputted a number.
7	Search	grep "bash" /etc/passwd | wc -l	Outputted "1", representing actual interactive users.
8	Users	whoami && groups	Outputted "nakama" and confirmed presence in the "sudo" group.
9	Users	sudo useradd -m analyst	Created the new user analyst along with a home directory.
10	Users	sudo passwd analyst && su analyst whoami exit	Set password, switched user (whoami showed analyst), then

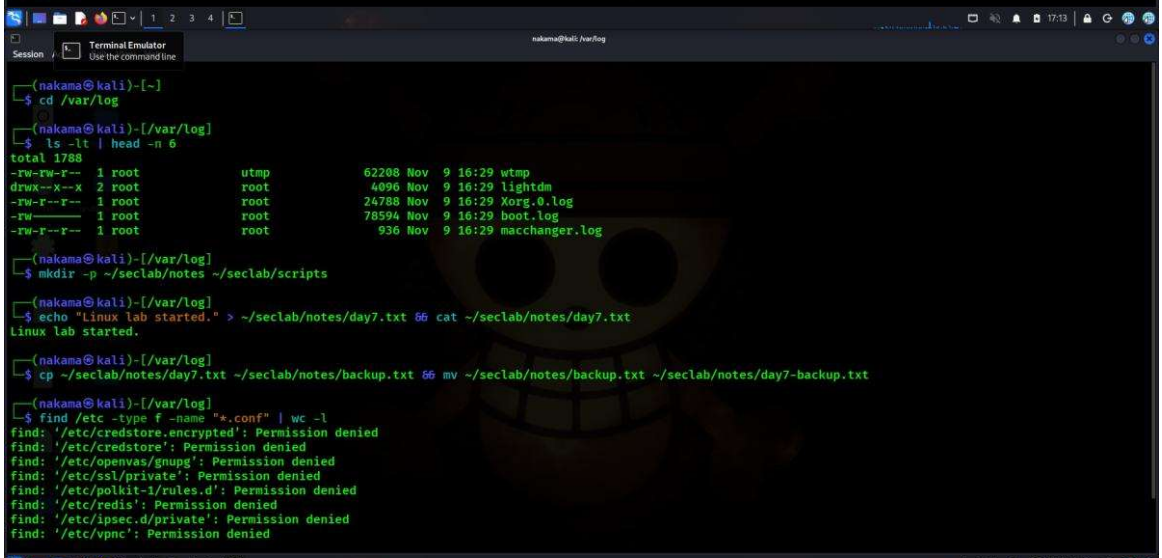
			returned to main shell.
11	Permissions	chmod 600 ~/seclab/notes/day7.txt && ls -l ~/seclab/notes/day7.txt	Showed permissions as -rw-------.
12	Permissions	echo "echo Hello from \$(whoami)" > ~/seclab/scripts/hello.sh && chmod +x ~/seclab/scripts/hello.sh && ~/seclab/scripts/hello.sh	Outputted "Hello from nakama".
13	Permissions	find / -perm -4000 -type f 2>/dev/null | wc -l	Found total SUIDs (~33)
14	Processes	ps aux | grep bash	Displayed process list and PID for the active bash session.
15	Processes	sleep 300 & ps kill <PID>	Started background task, found its PID, and terminated it successfully.
16	Logs	tail -n 20 /var/log/auth.log	Not found
17	Logs	grep -c "sudo" /var/log/auth.log	Auth.log not found
18	Network	sudo ss -tulpn	Nothing
19	Packages	sudo apt update && sudo apt install tree -y && tree ~/seclab	Updated repos, installed tree, and printed the visual structure of seclab folder.
20	Scripting	echo -e "#!/bin/bash\ndate\nwhoami\nwho | wc -l" > ~/seclab/scripts/check.sh && chmod +x ~/seclab/scripts/check.sh && ~/seclab/scripts/check.sh	Printed the current date, username (nakama), and "1" (number of logged-in users).

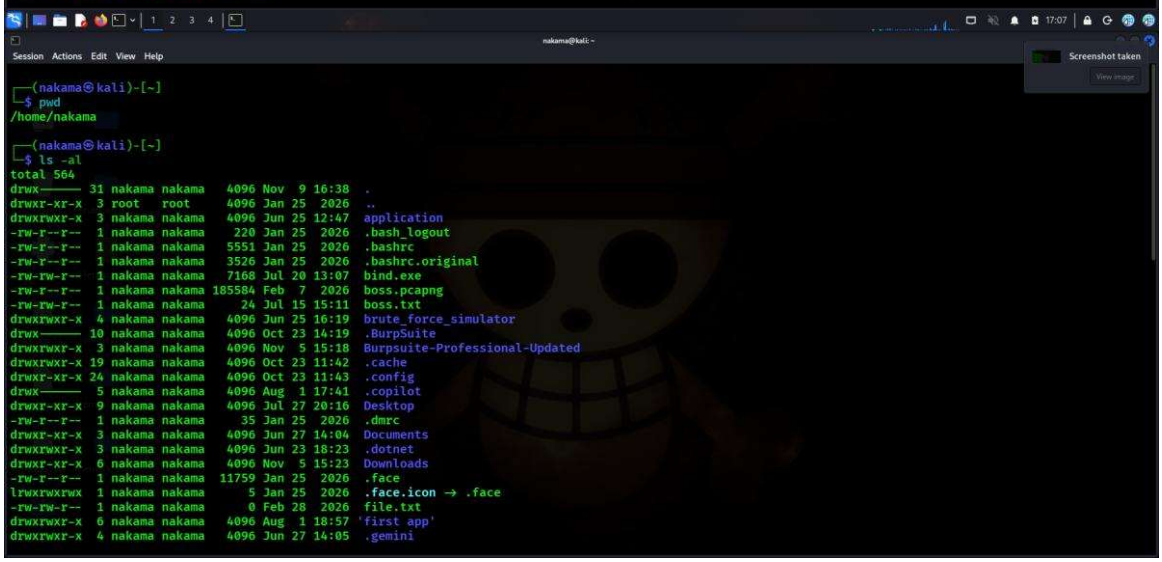
Snapshot of terminal:-


Terminal Emulator
Use the command line
Home
book_sum...
File System
pass.txt
Trash
brute_simu...
shell.exe
Metasploit...
unix_users1...
roadmap_g...
unix_passw...
process_kill
rockyou.txt
network_sif...
resume_help


Session Actions Edit View Help
┌──(nakama㉿kali)-[/var/log]
└─$ su analyst
Password:
$ whoami
analyst
$ exit
┌──(nakama㉿kali)-[/var/log]
└─$ chmod 600 ~/seclab/notes/day7.txt && ls -l ~/seclab/notes/day7.txt
-rw------- 1 nakama nakama 19 Nov 9 17:12 /home/nakama/seclab/notes/day7.txt
┌──(nakama㉿kali)-[/var/log]
└─$ echo "echo Hello from \$(whoami)" > ~/seclab/scripts/hello.sh && chmod +x ~/seclab/scripts/hello.sh && ~/seclab/scripts/hello.sh
Hello from nakama
┌──(nakama㉿kali)-[/var/log]
└─$ find / -perm -4000 -type f 2>/dev/null | wc -l
33
┌──(nakama㉿kali)-[/var/log]
└─$ ps aux | grep bash
nakama 29558 0.0 0.0 6556 2504 pts/0 S+ 17:26 0:00 grep --color=auto bash
┌──(nakama㉿kali)-[/var/log]
└─$ sleep 300 &
ps
kill <PID>
zsh: parse error near `\n'
┌──(nakama㉿kali)-[/var/log]
└─$


Text Editor
Simple Text Editor
Help
nakama@kali: /var/log
17:21
(nakama㉿kali)-[/var/log]
$ grep "bash" /etc/passwd | wc -l
1
(nakama㉿kali)-[/var/log]
$ whoami && groups
nakama
nakama adm dialout cdrom floppy sudo audio dip video plugdev users netdev scanner bluetooth lpadmin wireshark vboxsf kaboxer
(nakama㉿kali)-[/var/log]
$ sudo useradd -m analyst
[sudo] password for nakama:
(nakama㉿kali)-[/var/log]
$ sudo passwd analyst
New password:
Retype new password:
passwd: password updated successfully
(nakama㉿kali)-[/var/log]
$ su -analyst whoami
su: invalid option -- 'a'
Try 'su --help' for more information.
(nakama㉿kali)-[/var/log]
$ su analyst
Password:
su: Authentication failure
(nakama㉿kali)-[/var/log]
$ su analyst
Password:


Terminal Emulator
Use the command line
Session
nakama@kali: /var/log
17:13
(nakama㉿kali)-[~]
$ cd /var/log
(nakama㉿kali)-[/var/log]
$ ls -lt | head -n 6
total 1788
-rw-rw-r-- 1 root utmp 62208 Nov 9 16:29 wtmp
drwx--x--x 2 root root 4096 Nov 9 16:29 lightdm
-rw-r--r-- 1 root root 24788 Nov 9 16:29 Xorg.0.log
-rw------- 1 root root 78594 Nov 9 16:29 boot.log
-rw-r--r-- 1 root root 936 Nov 9 16:29 macchanger.log
(nakama㉿kali)-[/var/log]
$ mkdir -p ~/seclab/notes ~/seclab/scripts
(nakama㉿kali)-[/var/log]
$ echo "Linux lab started." > ~/seclab/notes/day7.txt && cat ~/seclab/notes/day7.txt
Linux lab started.
(nakama㉿kali)-[/var/log]
$ cp ~/seclab/notes/day7.txt ~/seclab/notes/backup.txt && mv ~/seclab/notes/backup.txt ~/seclab/notes/day7-backup.txt
(nakama㉿kali)-[/var/log]
$ find /etc -type f -name "*.conf" | wc -l
find: '/etc/credstore.encrypted': Permission denied
find: '/etc/credstore': Permission denied
find: '/etc/openvas/gnupg': Permission denied
find: '/etc/ssl/private': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/redis': Permission denied
find: '/etc/ipsec.d/private': Permission denied
find: '/etc/vpnc': Permission denied


Session Actions Edit View Help
nakama@kali: ~
17:07
Screenshot taken
View image
(nakama㉿kali)-[~]
$ pwd
/home/nakama
(nakama㉿kali)-[~]
$ ls -al
total 564
drwx------ 31 nakama nakama 4096 Nov 9 16:38 .
drwxr-xr-x 3 root root 4096 Jan 25 2026 ..
drwxrwxr-x 3 nakama nakama 4096 Jun 25 12:47 application
-rw-r--r-- 1 nakama nakama 220 Jan 25 2026 .bash_logout
-rw-r--r-- 1 nakama nakama 5551 Jan 25 2026 .bashrc
-rw-r--r-- 1 nakama nakama 3526 Jan 25 2026 .bashrc.original
-rw-rw-r-- 1 nakama nakama 7168 Jul 20 13:07 bind.exe
-rw-r--r-- 1 nakama nakama 185584 Feb 7 2026 boss.pcapng
-rw-rw-r-- 1 nakama nakama 24 Jul 15 15:11 boss.txt
drwxrwxr-x 4 nakama nakama 4096 Jun 25 16:19 brute_force_simulator
drwx------ 10 nakama nakama 4096 Oct 23 14:19 .BurpSuite
drwxrwxr-x 3 nakama nakama 4096 Nov 5 15:18 Burpsuite-Professional-Updated
drwxrwxr-x 19 nakama nakama 4096 Oct 23 11:42 .cache
drwxr-xr-x 24 nakama nakama 4096 Oct 23 11:43 .config
drwx------ 5 nakama nakama 4096 Aug 1 17:41 .copilot
drwxr-xr-x 9 nakama nakama 4096 Jul 27 20:16 Desktop
-rw-r--r-- 1 nakama nakama 35 Jan 25 2026 .dmrc
drwxr-xr-x 3 nakama nakama 4096 Jun 27 14:04 Documents
drwxrwxr-x 3 nakama nakama 4096 Jun 23 18:23 .dotnet
drwxr-xr-x 6 nakama nakama 4096 Nov 5 15:23 Downloads
-rw-r--r-- 1 nakama nakama 11759 Jan 25 2026 .face
lrwxrwxrwx 1 nakama nakama 5 Jan 25 2026 .face.icon → .face
-rw-rw-r-- 1 nakama nakama 0 Feb 28 2026 file.txt
drwxrwxr-x 6 nakama nakama 4096 Aug 1 18:57 'first app'
drwxrwxr-x 4 nakama nakama 4096 Jun 27 14:05 .gemini

Step 2: Investigate a suspicious server

Step	Command	What would worry you in the output?
Who	Who, whoami	Seeing an unknown username or multiple active TTY/PTS sessions from unknown IP addresses.
What	Ps aux,top (or htop)	A strangely named process (or generic name like "sysupdate") consuming 90%+ CPU, indicating a crypto miner.
Where	sudo ss -tupan (or netstat)	An ESTABLISHED connection on an unusual high port (like 4444 or 1337) communicating with an external Russian/foreign IP.
When	Var/log/auth.com	Hundreds of failed SSH root login attempts followed by a successful login at 3:00 AM.
How	history (or cat ~/.bash_history)	Commands showing wget downloading a malicious script, or discovering that the history file was deliberately cleared.
Fix	kill -9 <PID> && sudo ufw deny from <IP>	The malicious process immediately restarts itself with a new PID after being killed, showing it has persistence via a cronjob.

Step 3: Get ready for Day 8

Wireshark installed for Day 8: [✓] Done